# Concern Resolution Process for In-house Manufacturing

## QC Story

# QC STORY PROJECT

## PROJECT TITLE :-

**"To eliminate the concern of Water leakage through Garnish in Marazzo model"**

**Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.**

| | Plan | Actual |
|---|---|---|
| Start Date | **10.05.19** | **10.05.19** |
| Completion Date | **20.11.19** | **27.12.19** |

Current Status: **CLOSED**

# Cross Functional Team

| No | Description | Name | Designation | Dept | Phone |
|---|---|---|---|---|---|
| 1 | **Team Leader** | Abhishek Sonawane | L7-O | PRD | 9881751802 |
| 2 | **Mentor** | Chikhale Sandip | L5-M | STCF | 9922408138 |
| 3 | **Member 1** | Satish Pagar | L7-0 | MQA | 9423309400 |
| 4 | **Member 2** | GANESH BENDALE | L7-O | PRD | 9850952219 |
| 5 | **Member 3** | Pankaj Patil | L7-0 | SQA | 9689922687 |
| 6 | **Member 4** | Sachin Bhamare | L7-O | MQA | 9158960111 |

# Step1. Select Subject

**Brief description of Problem:-**

**To eliminate the concern of Water leakage through Garnish in Marazzo model**

What : Water leakage through Garnish in Marazzo Model.

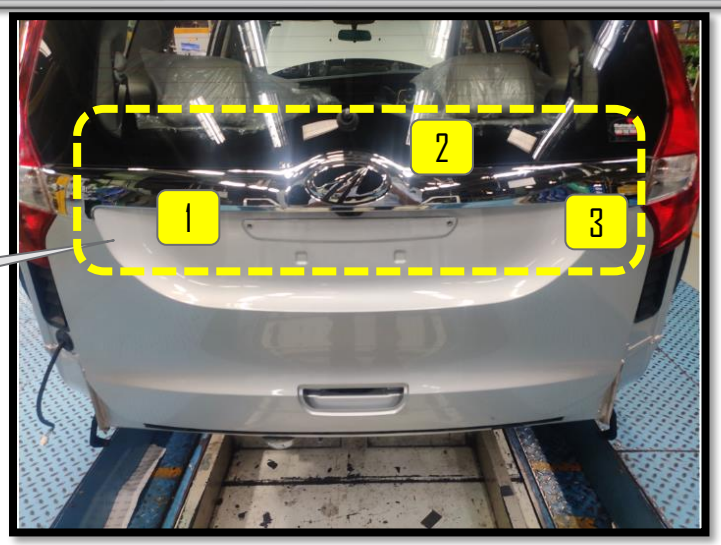
Where : At shower area.

When : Since Jan19

Who : By associate at shower stage.

WHY : High rework, FRC /FBO loss, Repeat issue at CAI-Shower test.

HOW : Avg. 14 cases/month reported at CAI-Shower Test.



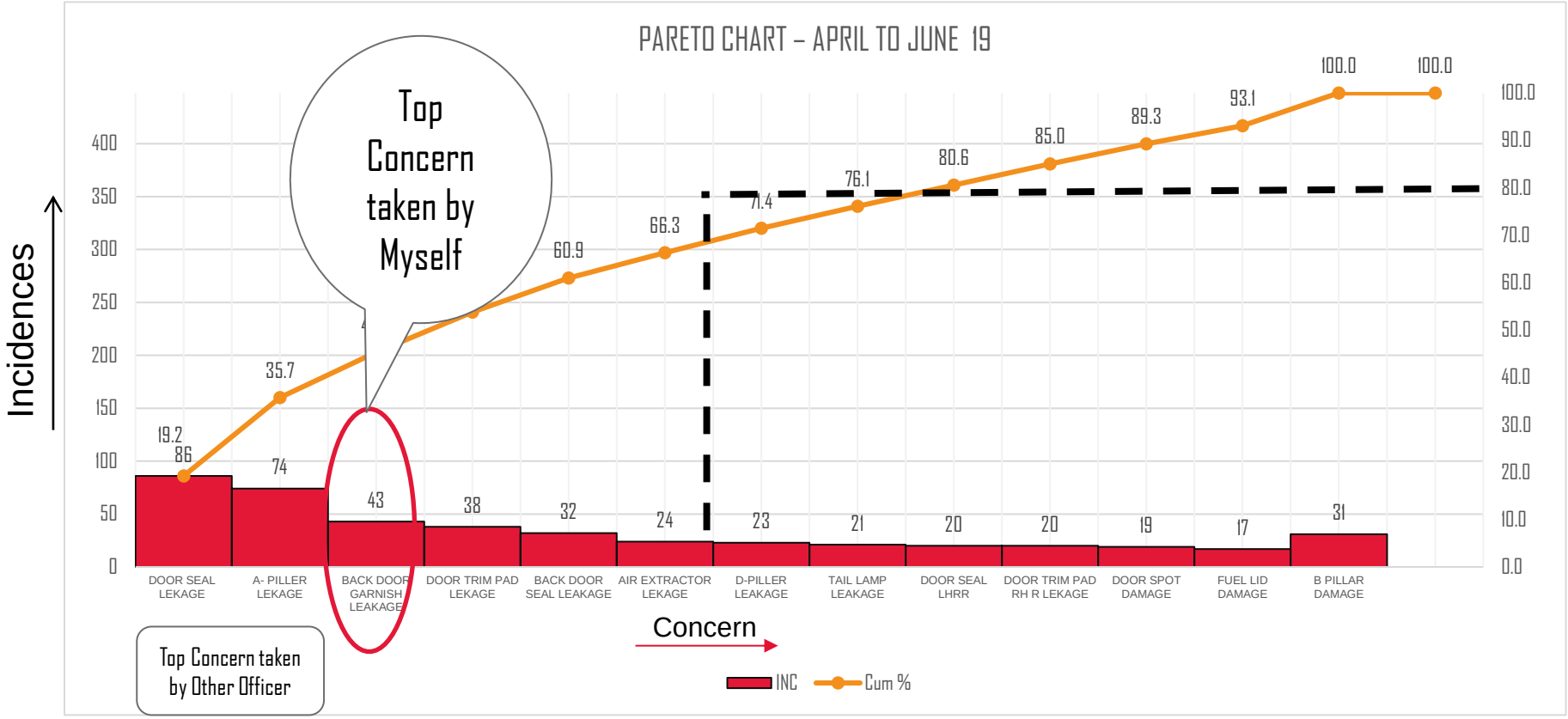

**WATER ENTRY SOURCE _WATER-LEKAGE THROUGH BACK DOOR GARNISH [MARAZZO]**

# 2. Justify the choice (reasons)

Crusade 2.0

**Departmental Objective :-**
**BIQ – DPR Improvement**
- **FRC / FBO Improvement**
- **Scrap Cost Reduction**

# 2. Justify the choice




Inference:
Water Leakage due to garnish fitment ,is one of the contributor and repeated chronic concern, hence taken for resolution

# 2. Justify the choice

## Severity & Consequences of Concern :-

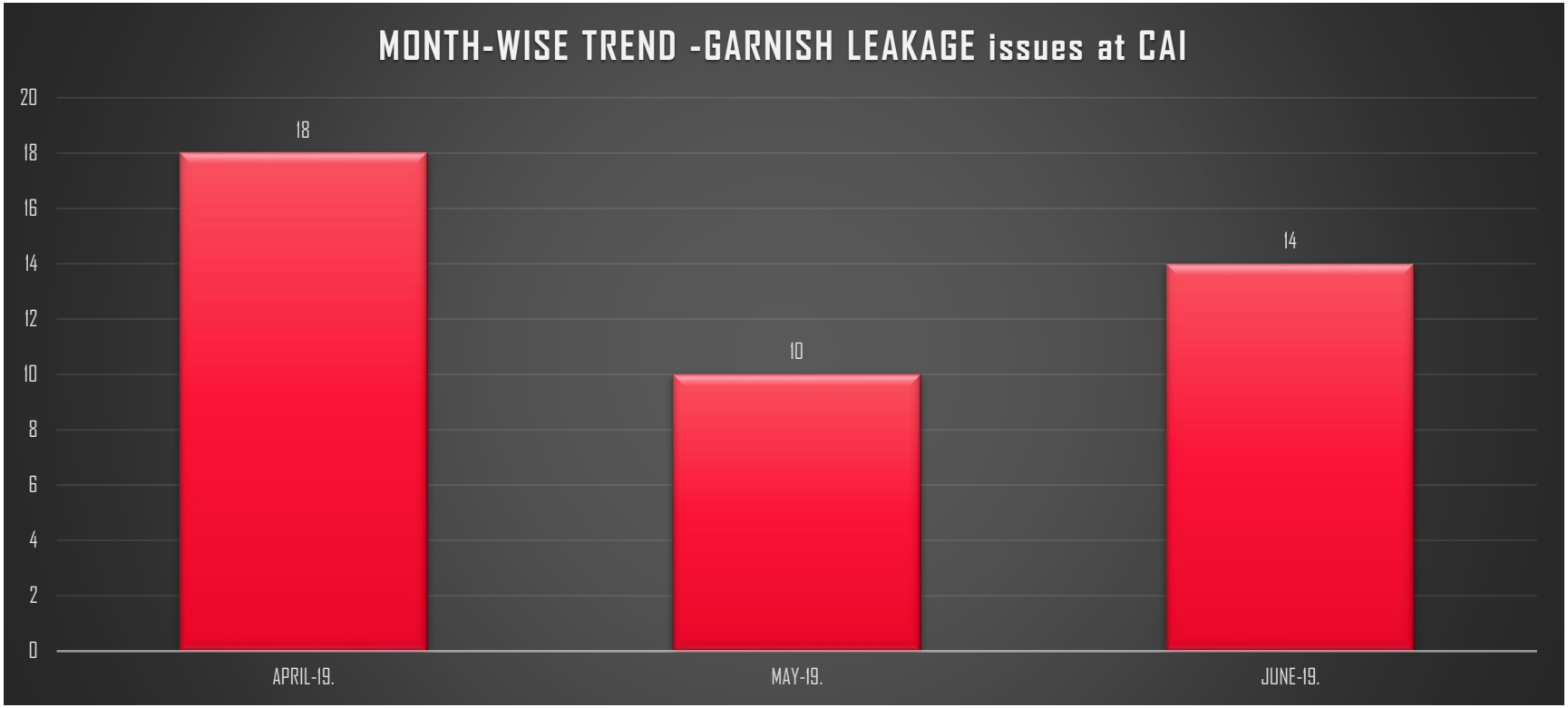
- Avg. 14 Cases reported per month, for rework, 4-5 V1 concern reported .
- Customer Dissatisfaction due to Water leakage.
- Rework Difficult
- Scrap cost for Garnish (Rs 1890 per Part.
- Rework time 20 min (approx.) for each vehicle by 1 operator.
- Rework cost is 50,940 Rs/year

## Cost of Poor Quality - Rework

| Rework Cost Estimation | | |
|---|---|---|
| Description | Units | |
| Rewok Time | Min | 20 |
| Avg Concerns/Day | Nos | 1 |
| Total Required Rework Time pe | Min | 20 |
| Effective Working time | MIn | 465 |
| Operator Require Per Day | Nos | 0.11 |
| Approximate Cost Per Day | Rs | 1390 |
| Operator Cost/ Concern | Rs | 101.88 |
| Avg no of working days | Days | 25 |
| Operator Cost Per Month | Rs | 2547.1 |
| Operator Cost /Year | Rs | 50,940 |

# 2. Justify the choice

## MONTHWISE DEFECT TREAD :-




Incidences

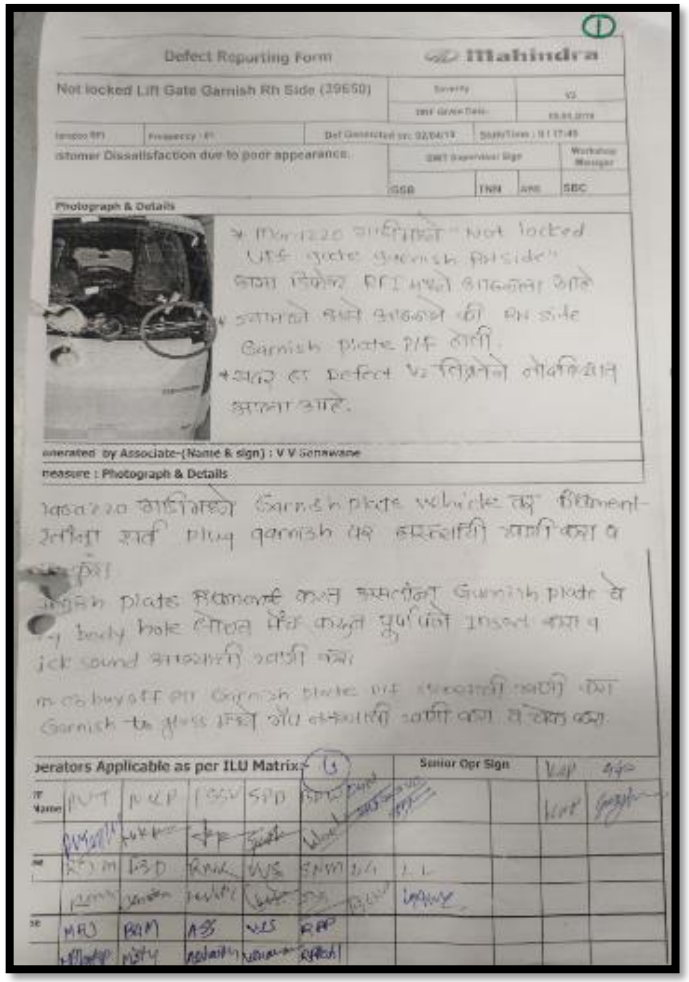
Month

Inference : Average 14 No.s concerns reported per month from April19 to June-19

# 2.b Immediate Containment Action

- Training for stage associates, explaining main step and key points.
- Emphasis on Garnish Fitment main steps and keypoints.
- 100% lot checked for supplier variability.
- DRF given to all stage associates & buy off associates, .

DRF TO ALL STAGE ASSOCAITES



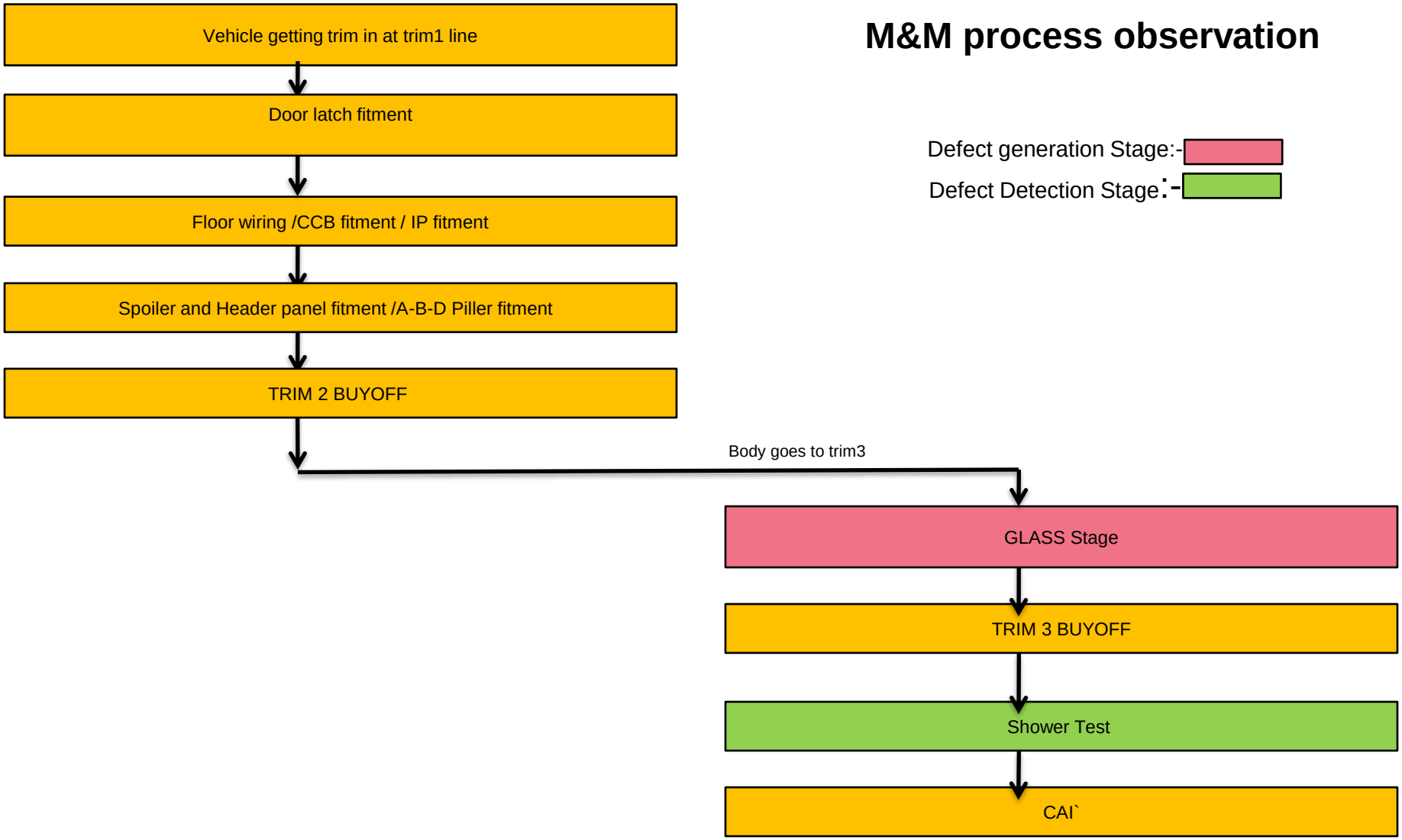
# 3. Understand the current situation

## Flow Chart :- Observation of Process

### M&M process observation

Defect generation Stage:-

Defect Detection Stage:-



# Step 3. Understanding Current Situation

## Observation of process

Body from door stage

Garnish Kept as Per Model in rack

Selecting Garnish as per model from Rack

Fitment of Back-Door Garnish

Checkpoint at Trim3 Buyoff

CHECKING AT CAI-SHOWER TEST

Garnish Fitment at Glass stage

Defect generation stage
Defect identification stage

# Step 3. Understanding Current Situation

**4M Analysis (Part 1)**
**(Conformity Analysis - 'L0')**

PU/ Area :SCORPIO/XYLO TCF
Cell / BWT / Line: BWT-6/TRIM3

Requested By- MQA | 12.04.2019 | Source -Marazzo RF | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : **Not locked Lift Gate Garnish Rh Side.**

FMEA Severity /Gravity : ☐ V1+ ☐ V1 ☑ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.-39650 | Model : Marazzo

Date of Incidence: Assy : 02.04.19 | Shift / Time : 2nd / 17: 48 | Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☐ SC

**Immediate containment action for Customer Protection :**
YARD CHECK DONE
DRF GIVEN TO ALL STAGE ASSOCAITE

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. / QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - BWT . :

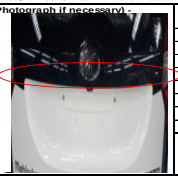
Does a control already exist ? ☐ YES ☑ NO

**Stage / Post Analysis -**

| Man( Human Action) | | Method( process) | | Other | |
|---|---|---|---|---|---|
| Operator Name : V V Sonawane | | | | | |
| Is operator trained ? | ☑ YES ☐ NO | Main steps are respected ? | ☐ YES ☑ NO | Is there a Control Plan ? (Record in Analysis table) | ☑ YES ☐ NO |
| Operator level (LL.U) | I | Is there a related key point in SOP / SOS? | ☑ YES ☐ NO | Is the packaging standard defined for the part ? | ☑ YES ☐ NO |
| OR Dexterity level(1…4) | L2 | Are the reasons for key point know n ? | ☑ YES ☐ NO | Is the packaging standard respected ? | ☑ YES ☐ NO |
| Is operator under training ? | ☐ YES ☑ NO | Are operating reqmnts respected ? | ☑ YES ☐ NO | Is there any alert mechanism for failure ? | ☑ YES ☐ NO |
| Is operator temporary for Stage ? | ☐ YES ☑ NO | Is the defect part of the Buy off / Checkman list ? | ☑ YES ☐ NO | N/A | |
| Bad rew ork ? | ☐ YES ☑ NO | Was it detected by Inspector/ Buoff Opr ? | ☐ YES ☑ NO | | |
| Is the specified /Tool / Fixture facility used ? | ☑ YES ☐ NO | In only one s | A V Tambe ☐ YES ☐ NO | Is the specified product used ? | ☑ YES ☐ NO |
| Last calibration dates | | In only one line ? | ☑ YES ☐ NO | Did it evolve / Change ? | ☑ YES ☐ NO |
| Facility /Tool / Fixture in good condition ? | ☑ YES ☐ NO | The problem follow s the vehicle ? | ☐ YES ☑ NO | Is the storage SOP respected ? | ☑ YES ☐ NO |
| Is it a substitute facility ? | ☐ YES ☑ NO | The problem follow s the part ? | ☐ YES ☑ NO | Is the shelf life / expiry date respected ? | ☐ YES ☐ NO |
| Are machine parameters known ? (Record in Analysis table) | ☑ YES ☐ NO | Is it a repeat problem ? | ☐ YES ☑ NO | Is the part as per FIFO? | ☑ YES ☐ NO |
| Parameters are respected ? | ☑ YES ☐ NO | Is the suspected parameter measurable ? (Record in Analysis table) | ☐ N/A ☐ NO | | |
| **Machine** | | **Product Failure** | | **Material/ Consumable** | |

*Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)*

**Analysis Conclusion (Attach Sketch/ Photograph if necessary) -**
1. Not locked Lift Gate Garnish Rh Side is observed at Marazoo RFI.
2. There is no evidance found in traveller card for rework.
3. Associates mistake, not press the garnish fulley.
5. Buy off stage checkman not able to detected the defect.



| Sno | Product / Process / Machine / Parameter | Spec. | Actual | Remark |
|---|---|---|---|---|
| | NIL | | | |

Not Applicable

**Is 4M Analysis Part 2 required ?** ☐ YES ☑ NO

**Is the substitute transport mode used ?** ☐ YES ☑ NO | **Is the packaging defined ?** ☐ YES ☐ NO

**Actions -**

| 1.Containment Action | Resp | Date | Status |
|---|---|---|---|
| Awareness given to reworkmen and chekman ( DRF given) | DDW/TN | 11.04.19 | Closed |

| 2. Permanent Action - | Resp | Date | Status |
|---|---|---|---|
| check point added by pulling at Trim3 Buyoff.L1 required | tnn | 12.04.2019 | Open |

Analysed by Name/s - T Nandy
Date - 11.04.2019

Choice : ☑ WSS ☐ QC Story

Inference from L0
Problem follows part & vehicle
Problem is a repeat problem
L1 analysis required

| 4M analysis Good and Not so good observations | |
|---|---|
| MAN | Operator dext. Level L3 |
| Method | Operator aware of key point and check present at buyoff |
| Others | Control plan adhered on stage |
| Machine | Tools used are calibrated |
| Product failure | Problem follows part and vehicle |
| Material | Handling and shelf life of part adhered |

**NO QUICK WIN**

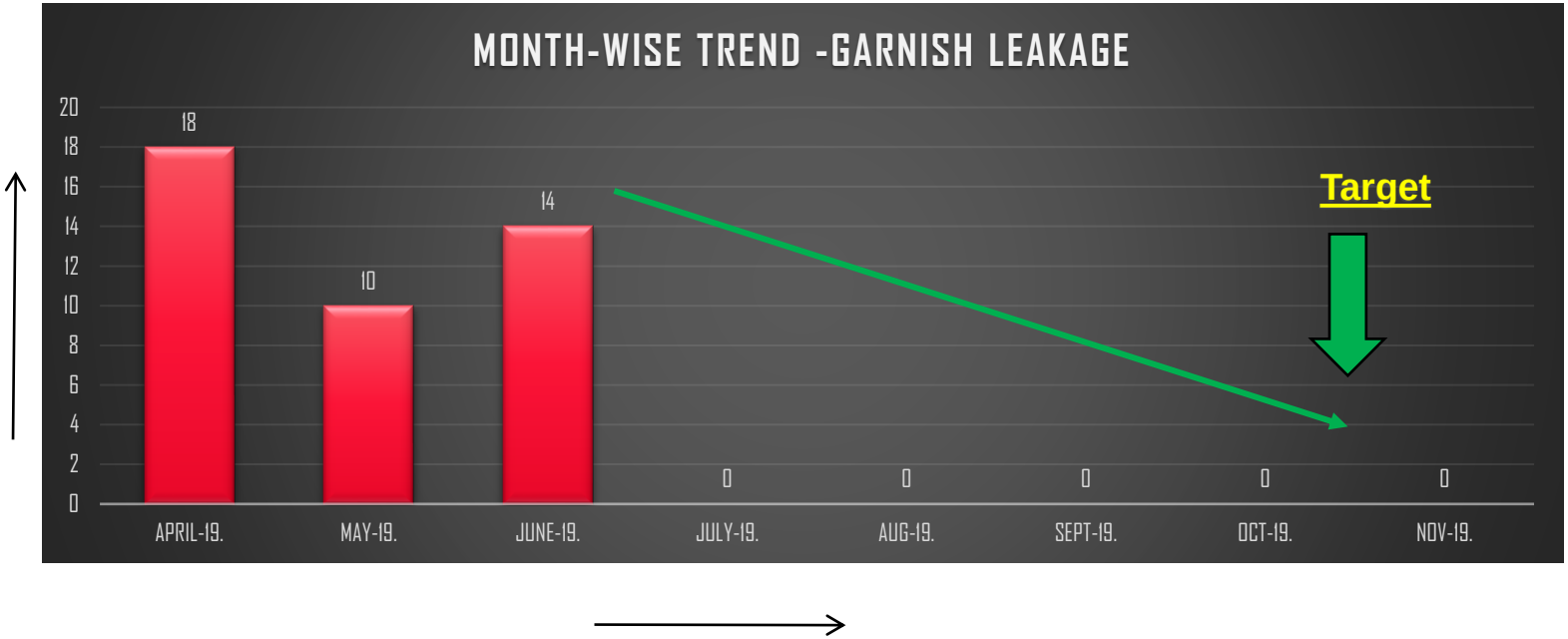
# Step 3. Understanding Current Situation

## Good & Not so good observations

- Issue observed only in Marazzo Model.
- Issue reported in  both the shifts
- Repair time is high, around 20 minutes per vehicle
- Level 3 operators are deployed for Garnish Fitment
- Operators are aware of Main Step, key point & check point at respective stages.
- Garnish is getting poor fit in Marazzo vehicles.

# 4. Select targets

| | |
|---|---|
| Present Level: | **Avg. 14 Incidences /month are being reported at CAI** |
| Target Level: | **Water leakage through Garnish in Marazzo model by Nov-19.** |



# Step 5. Analysis

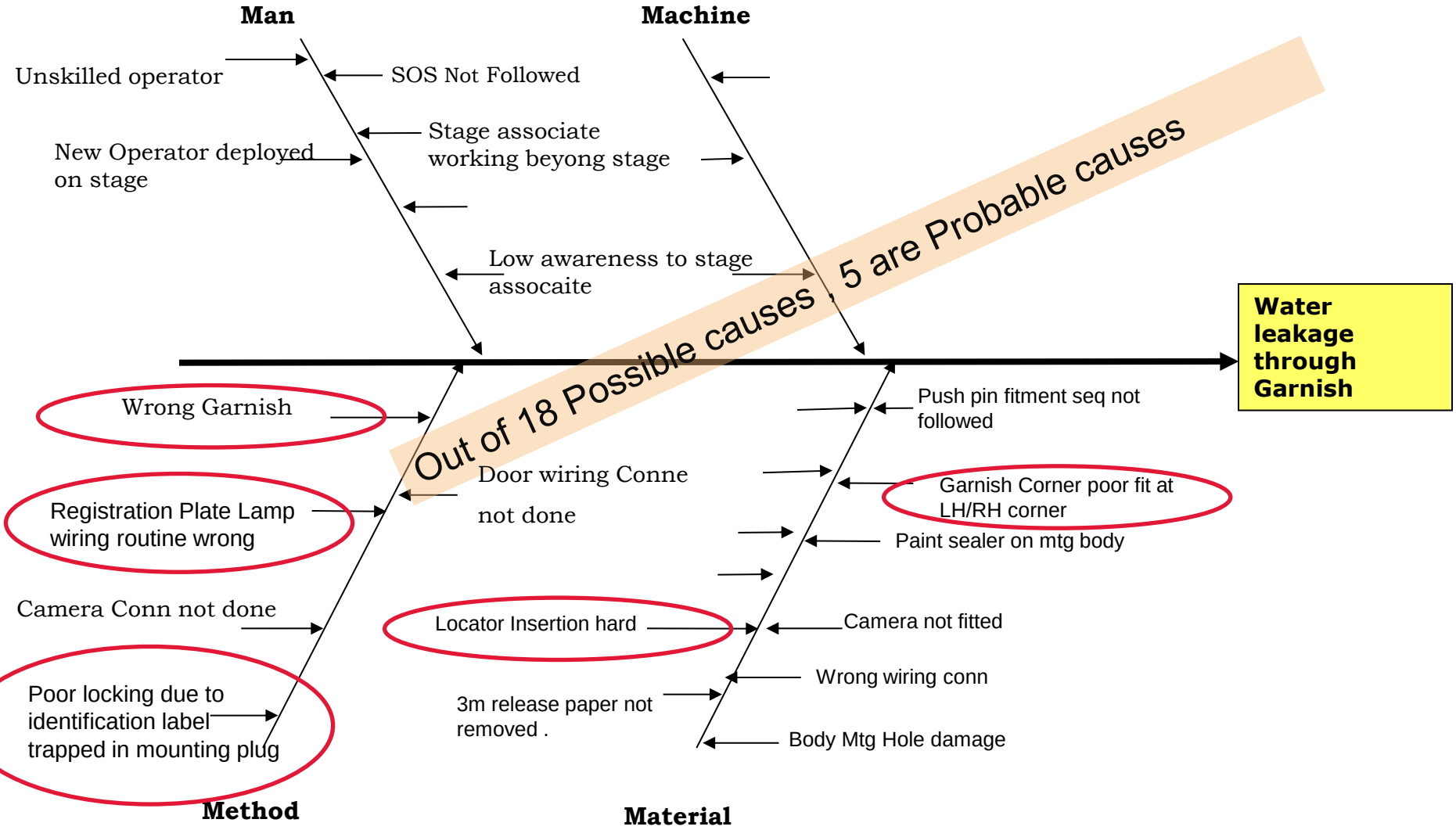
## List of Possible causes identified in Brain storming session

**Brainstorming Done along with CFT & we have found out 18 Possible Causes for this defect.**

| S N | Probable Cause |
|---|---|
| | Garnish Fitment ON MARAZZO MODEL |
| 1 | Unskilled operator |
| 2 | New Operator deployed on stage |
| 3 | SOS Not Followed |
| 4 | Wrong Garnish Fitment |
| 5 | Body mtg hole damage |
| 6 | Locator Insertion hard |
| 7 | SOS inadequate |
| 8 | 3m release paper not removed |
| 9 | Garnish corner poor fit at LH/RH Corner |
| 10 | Paint sealer on body mtg hole |
| 11 | push-pin fitment sequence not followed |
| 12 | Camera Not fitted |
| 13 | Camera Connection not done. |
| 14 | Door wiring Conn not done |
| 15 | Wrong wiring done from trim 1. |
| 16 | Registration plate lamp bulb loose |
| 17 | Registration Plate Lamp wiring routine wrong |
| 18 | Poor locking due to identification label trapped in mouting plug |

# Step 5. Analysis

## 5.1 Identify possible causes: Fish Bone diagram



# Step 5. Analysis

List of Probable causes :

| S N | Probable Cause | Ganesh | Mahesh | Nandy | Sakha | Satish | Santosh | Total |
|---|---|---|---|---|---|---|---|---|
| | **Garnish Fitment** | | | | | | | |
| 1 | Unskilled operator | 0 | 0 | 0 | 0 | 1 | 1 | 2 |
| 2 | New Operator deployed on stage | 3 | 1 | 3 | 1 | 3 | 1 | 12 |
| 3 | SOS Not Followed | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 4 | Wrong Garnish Fitment | 9 | 3 | 9 | 1 | 3 | 9 | 34 |
| 5 | Body mtg hole damage | 1 | 0 | 0 | 1 | 0 | 0 | 2 |
| 6 | Locator Insertion hard | 9 | 9 | 9 | 3 | 3 | 3 | 36 |
| 7 | SOS inadequate | 1 | 0 | 0 | 1 | 0 | 0 | 2 |
| 8 | 3m release paper not removed | 0 | 0 | 0 | 1 | 0 | 1 | 2 |
| 9 | Garnish poor fit at LH/RH Corner | 0 | 1 | 3 | 3 | 9 | 9 | 25 |
| 10 | Paint sealer on body mtg hole | 0 | 1 | 1 | 3 | 0 | 1 | 6 |
| 11 | push-pin fitmetn sequence not followed | 3 | 5 | 3 | 3 | 3 | 1 | 18 |
| 12 | Camera Not fitted | 1 | 0 | 0 | 1 | 0 | 0 | 2 |
| 13 | Connection not done. | 0 | 0 | 0 | 1 | 0 | 1 | 2 |
| 14 | Door Wiring Conn not done. | 1 | 0 | 0 | 1 | 0 | 3 | 6 |
| 15 | Wrong wiring done from trim 1. | 0 | 1 | 1 | 1 | 0 | 3 | 6 |
| 16 | Registration plate lamp bulb loose | 3 | 5 | 3 | 0 | 3 | 0 | 14 |
| 17 | Registration Plate Lamp wiring routine wrong | 9 | 9 | 9 | 3 | 3 | 3 | 36 |
| 18 | Poor locking due to identification label trapped in mouting plug | 3 | 3 | 3 | 9 | 9 | 9 | 36 |

**Selection Criteria**: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
**Threshold Value for selecting probable cause is 25.**

# Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering.

| Sr.No | Probable Cause | Reasons |
|---|---|---|
| 1 | Unskilled operator | All Trained assocaites |
| 2 | New Operator deployed on stage | All Trained assocaites |
| 3 | SOS Not Followed | SOS are followed |
| 5 | Body mtg hole damage | Hole as Dimension . |
| 6 | Locator Insertion hard | 50 parts Fitment Checked Found ok |
| 7 | SOS inadequate | SOS Verified found ok |
| 8 | 3m release paper not removed | line side stock checked for release paper missing. |
| 9 | Paint sealer on body mtg hole | 100 Bodies checked paint sealer found ok. |
| 10 | push-pin fitmetn sequence not followed | SOS Followed for Garnish Fitment |
| 11 | Camera Not fitted | Camera Fitment Poosible & found ok. |
| 12 | Connection not done. | Checked Click Sound of Socket Found Ok. |
| 13 | Wrong wiring done from trim 1. | 100 Veh checked wiring found ok as per model |
| 14 | Registration plate lamp bulb loose | Line Side Stock Checked for bulb loose found ok |

# 5.1 Identify Probable cause

| S N | Probable Cause |
|---|---|
| 1 | Wrong garnish Fitment |
| 2 | Locator Insertion Hard |
| 3 | Garnish Poor fit at LH/RH Corner. |
| 4 | Registration Plate Lamp wiring routine wrong |
| 5 | Poor locking due to identification label trapped in mounting plug |

# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing method | Testing and Observations | Conclusion |
|---|---|---|---|---|
| 1 | Wrong garnish Fitment | Visual check | It was found that, at Shower Test, total=4veh.,reported for Water leakage. After detail analysis ,it was found Back Door Garnish was replaced as it was wrong fitted on vehicle. While replacing it was poor fit by stage associate, and hence water leakage was observed. | Hypothesis valid |

BACK DOOR GARNISH FITMENT

**Contributors :-**

Model Name Written on lh corner of rack.

Garnish Stored in rack as per Model.

| Types Of Garnish | | |
|---|---|---|
| Sr.No | Model | Variant |
| 1 | M2 & M4 | Black Color |
| 2 | M6 | Silver Chrome |
| 3 | M8 | Silver Chrome with Camera Hole |

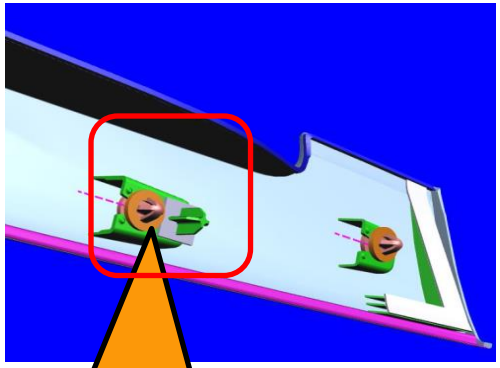
LINE-SIDE SORAGE BACK DOOR GARNISH

# TEST OF HYPOTHESIS

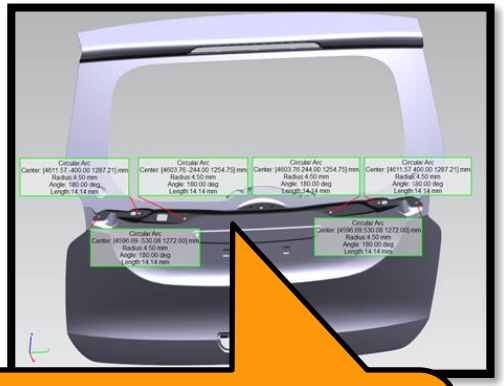
## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing method | Testing and Observations | Conclusion |
|---|---|---|---|---|
| 2 | Locator Insertion Hard | Visual check | Total =30 parts checked for locator. It was found that plug position was found ok. | Hypothesis invalid |

BACKDOOR GARNISH FITMENT (LOCATOR INSERTION DATA)



For Plug position, existing stock checked.
Dog house plug position was found ok.



Body Hole & garnish plug checked found ok.

Conclusion :
1.Actual 30 garnish checked for plug coming out from dog house.all part found ok.
2.15 Bodies checked for 6 holes all found within the specified dimension
3. Hypothesis is inValid.

# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing method | Testing and Observations | Conclusion |
|---|---|---|---|---|
| 3 | Garnish Poor Fit at LH/RH Corner | Visual check | Total=25veh checked, Process checked for garnish corner lh & rh fitment . It was observed that, leakage from LH & RH corner | **Hypothesis Invalid** |

Stage associate is working as per SOS, and doing fitment as sequence shown

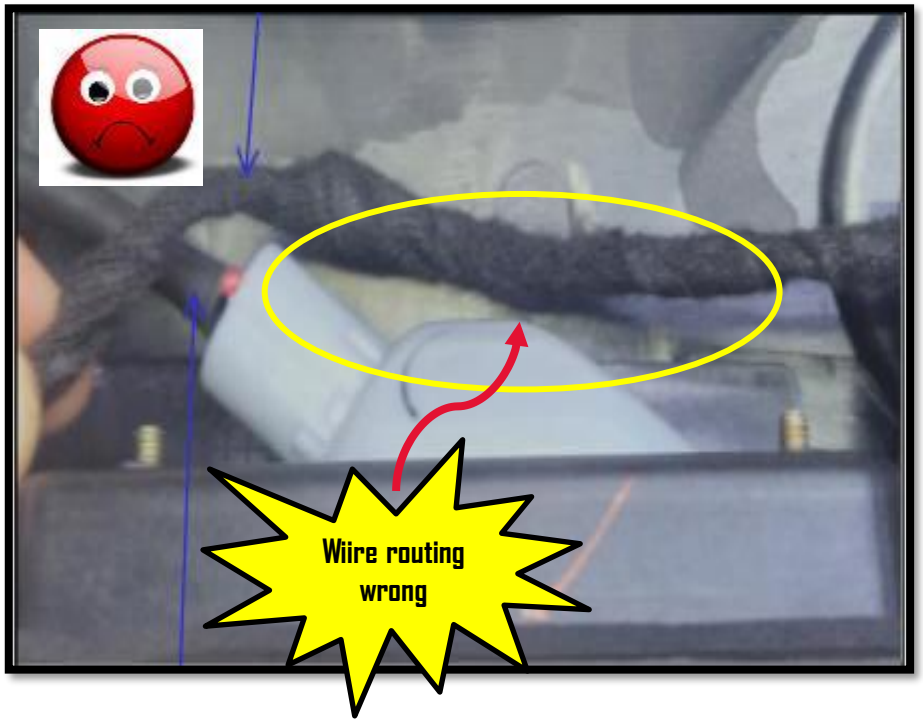
| Sr.No | Vin Number | Model | Garnish poor Fit at Corner LH/RH |
|---|---|---|---|
| 1 | MA1WA2ES1K2D39148 | M2 | OK |
| 2 | MA1WA2ES1K2D39146 | M2 | OK |
| 3 | MA1WA2ES1K2D39149 | M2 | OK |
| 4 | MA1WA2ES1K2D39150 | M2 | OK |
| 5 | MA1WA2ES1K2D39152 | M2 | OK |
| 6 | MA1WA2ES1K2D39153 | M2 | LH |
| 7 | MA1WA2ES1K2D39154 | M2 | LH |
| 8 | MA1WA2ES1K2D39157 | M8 | OK |
| 9 | MA1WA2ES1K2D39156 | M2 | RH |
| 10 | MA1WA2ES1K2D39146 | M2 | RH |
| 11 | MA1WA2ES1K2D39158 | M2 | RH |
| 12 | MA1WA2ES1K2D39161 | M2 | OK |
| 13 | MA1WA2ES1K2D39148 | M2 | OK |
| 14 | MA1WA2ES1K2D39162 | M2 | OK |
| 15 | MA1WA2ES1K2D39149 | M2 | OK |
| 16 | MA1WA2ES1K2D39150 | M2 | OK |
| 17 | MA1WA2ES1K2D39163 | M2 | OK |
| 18 | MA1WA2ES1K2D39164 | M2 | OK |
| 19 | MA1WA2ES1K2D39152 | M2 | OK |
| 20 | MA1WA2ES1K2D39153 | M2 | OK |
| 21 | MA1WA2ES1K2D39165 | M2 | LH |
| 22 | MA1WA2ES1K2D39167 | M2 | RH |
| 23 | MA1WA2ES1K2D39154 | M2 | OK |
| 24 | MA1WA2ES1K2D39168 | M4 | LH |
| 25 | MA1WA2ES1K2D39146 | M2 | LH |

**Conclusion –**
**Stage associate followed the SOS correctly & ensures correct fitment process.**
**Till 9 Vehicles reported at buy off for LH & RH Corner these results into leakage from garnish**

# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing method | Testing and Observations | Conclusion |
|---|---|---|---|---|
| 4 | Registration Plate Lamp wiring routine wrong | Visual check | 25 Garnish Checked for Wire routing. It was found that wirerouting was above socket. Total 25 veh checked, 10 veh found not ok. | Hypothesis Valid |




Conclusion :

1.At Gemba,during garnish locator fitment wire came near by locator so locator not inserted fully in body hole leads to garnish poor fitment &hence shower leakage .

2.It was found that, wire is getting entrapped with garnish and hence garnish plug is not getting locked.

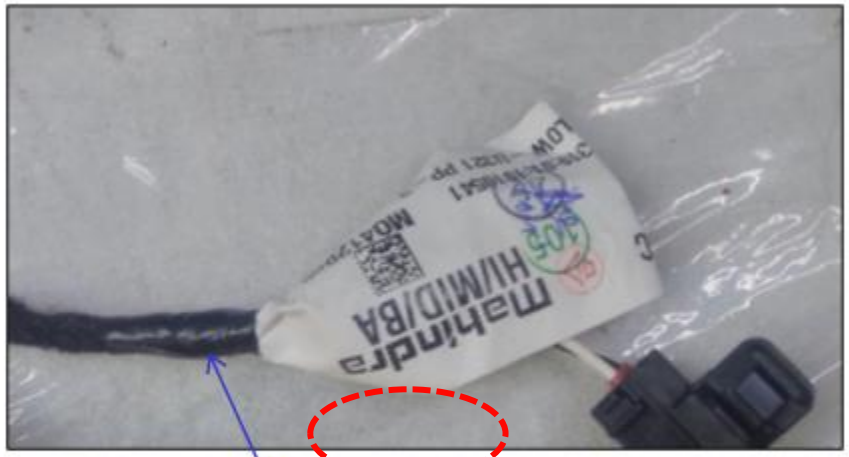
3. Hypothesis is Valid.

# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing method | Testing and Observations | Conclusion |
|---|---|---|---|---|
| 5 | Poor locking due to identification label trapped in mounting plug | Visual check | It was found that, Back door garnish is getting poor locked,as identification label is getting trapped while fitment. | Hypothesis valid |

### STUDY OF BACK-DOOR GARNISH




Identification tag bind only one side by black tape

**ANALYSIS:**

1. **It was found that, identification label was in loose condition.**
2. **This label is getting entrapped into mounting.**
3. **Hence plug is not getting locked.**
4. **This leads to Poor locking of Garnish,and hence Water leakage**

Identification label overlap with Back door garnish plug.

# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| Sr. No | Probable Cause | Testing method | Conclusion |
|---|---|---|---|
| 1 | Wrong garnish Fitment | Visual Check | Hypothesis valid |
| 2 | Locator Insertion Hard | Visual check | Hypothesis Invalid |
| 3 | Garnish Poor fit at LH/RH Corner. | Visual Check | Hypothesis invalid |
| 4 | Registration Plate Lamp wiring routine wrong | Visual check | Hypothesis valid |
| 5 | Poor locking due to identification label trapped in mounting plug | Visual Check | Hypothesis valid |

3 Valid Causes

# WHY - WHY Analysis

## 5.3 Identify the root causes:

| No | Valid Probable Cause | WHY | WHY | WHY | ACTION / COUNTER MEASURE |
|---|---|---|---|---|---|
| 1 | Wrong garnish Fitment | Associate Select wrong garnish | Garnish Kept as per model in rack. | No Sequencing of Garnish . | Sequencing Of Garnish |
| 2 | Registration Plate Lamp wiring routine wrong | Wire routing coming from supplier | Supplier not aware about the this failure mode | No specific requirement given to supplier | Requirement to be given to supplier |
| 3 | Poor locking due to identification label trapped in mounting plug | Identification label position near Back Door Garnish mtg. plug. | While fitment, identification Label gets overlap with mtg. plug. | Sticker position & size is incorrect | Correct the size and position |

# 6. Implement the corrective measures

## Possible Remedies Identified

**Legends**

↑ High ↓ Low

| S N | Root Cause | Remedy Developed | Time to implement | Cost of Improvement | Easily reversible | Complexity | Decision |
|---|---|---|---|---|---|---|---|
| 1 | No Sequencing of Garnish . | Sequencing Of Garnish | Low ↓ | Low ↓ | Yes | Low ↓ | Accepted |
| 2 | No specific requirement given to supplier | Requirement to be given to supplier | High ↑ | Low ↓ | No | Low ↓ | Accepted |
| 3 | Sticker position & size is incorrect | Correct the size and position | Low ↓ | Low ↓ | Yes | Low ↓ | Accepted |

# 6. Implement the corrective measures

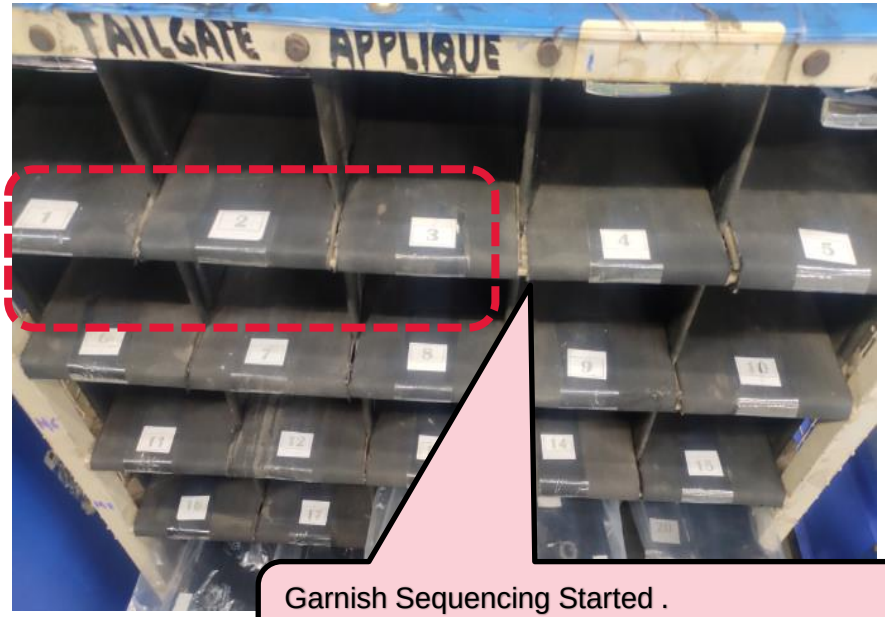
## 6.1 Actions Decided

| No | Root Cause | Action plan | Actual date | Responsibility | Status |
|---|---|---|---|---|---|
| 1 | Wrong garnish Fitment | Sequencing Of Garnish on rack started | 05.07.19 | Avinash Mahale | Closed |

**BEFORE**

No Sequencing of Garnish

**AFTER**

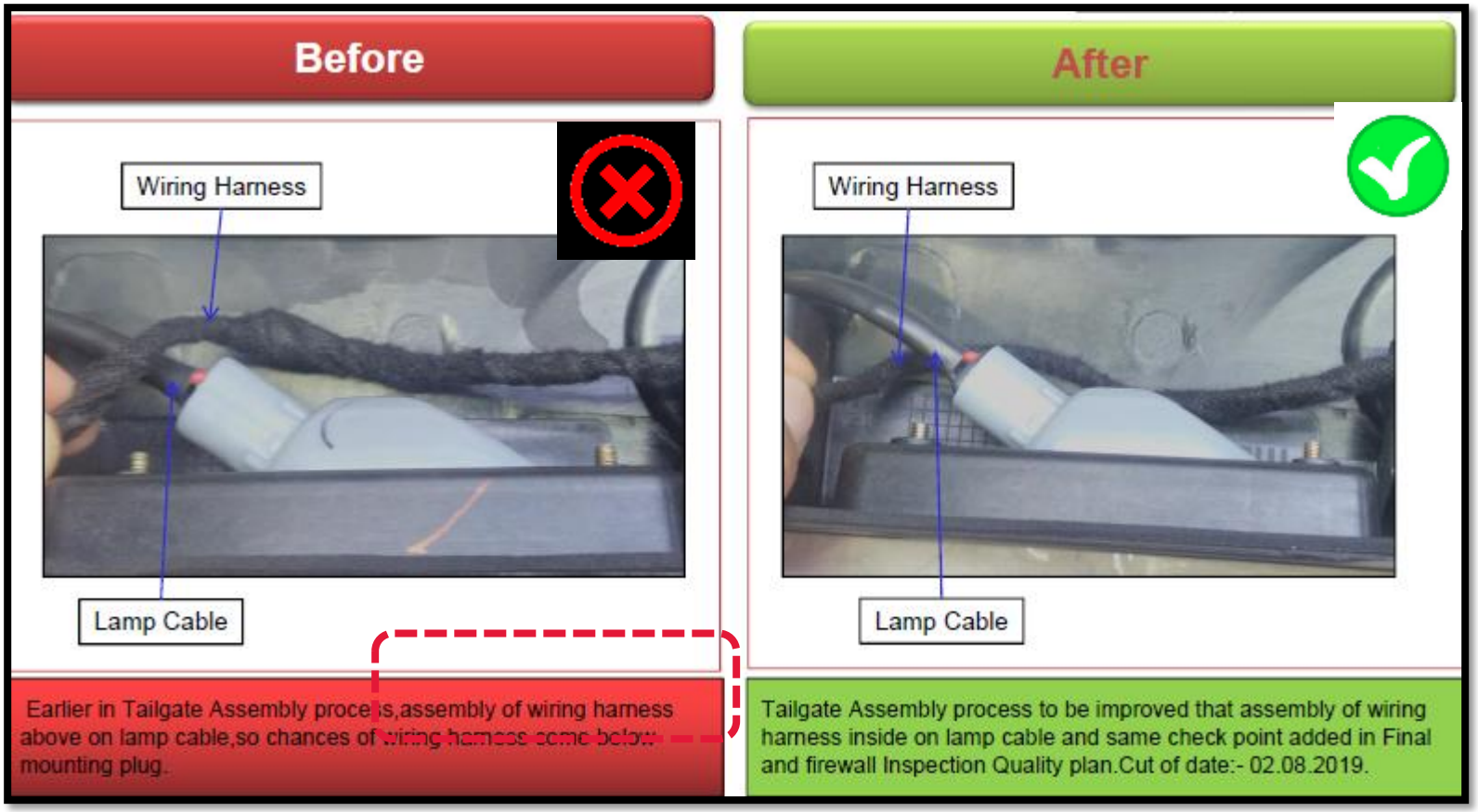



Garnish Sequencing Started .
Numbering On rack Done for the sequential feeding of garnish

# 6. Implement the corrective measures

## 6.1 Actions Decided

| No | Root Cause | Action plan | Actual date | Responsibility | Status |
|---|---|---|---|---|---|
| 2 | Registration Plate Lamp wiring routine wrong | Wiring harness cable asembly inside of lamp cable. | 02.08.19 | Pankaj Patil | Closed |



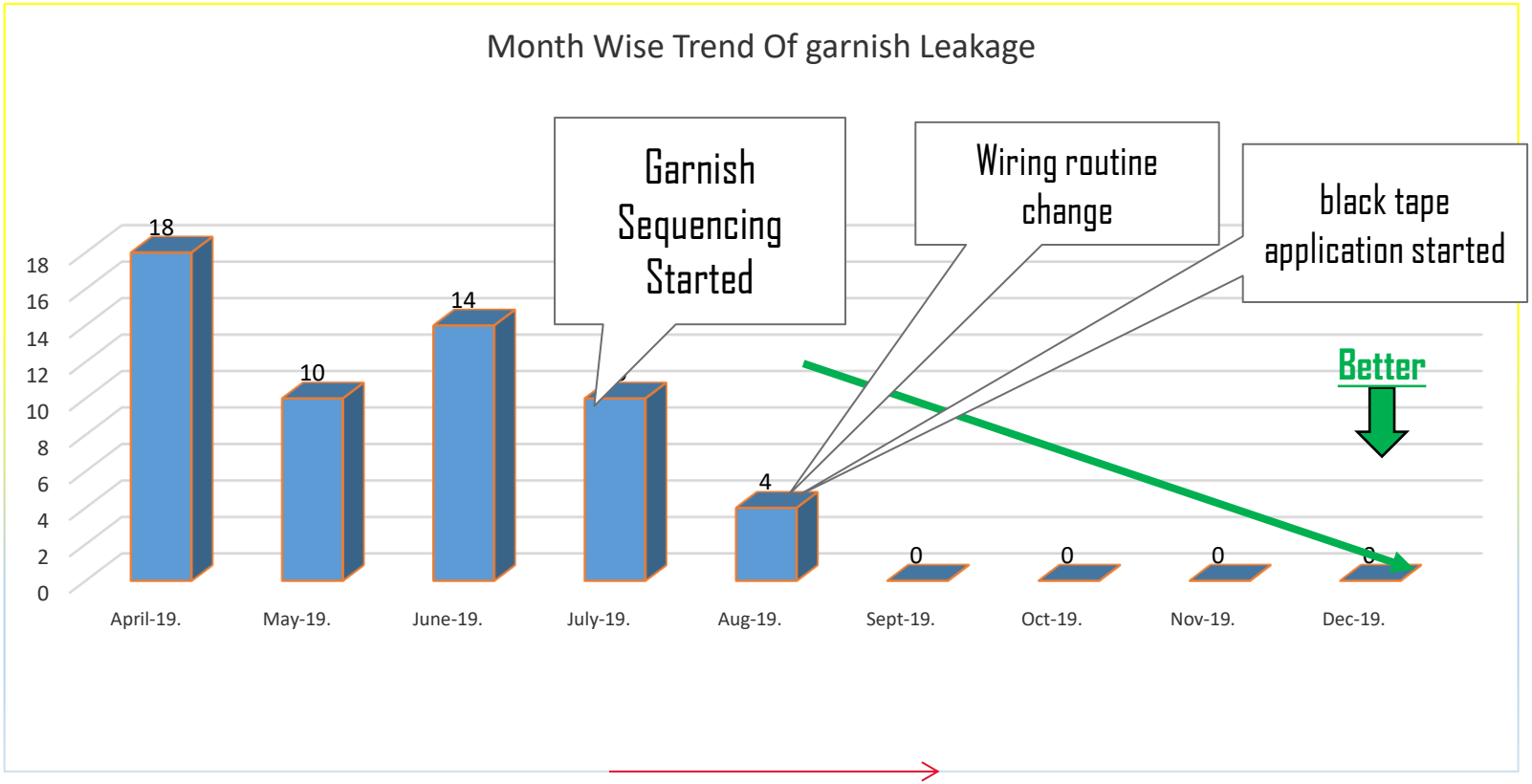
# 6. Implement the corrective measures

## 6.1 Actions Decided

| No | Root Cause | Action plan | Actual date | Responsibility | Status |
|---|---|---|---|---|---|
| 3 | Poor locking due to identification label trapped in mounting plug | Now in wiring harness ,identification tag bind on both side by black tape. | 2.8.19 | Pankaj Patil | Closed |

# 7. Check & confirm the results & effects

## 7.1 Verify Results:




**Inference – Concern not reported from after implementing the PCA actions.**

# 7. Check & confirm the results & effects

## 7.2 Full Scale Implementation plan:

| S.N | Action Taken | Actual date | Responsibility | Status |
|---|---|---|---|---|
| 1 | Garnish Sequencing Started | 05.07.19 | Avinash Mahale | CLOSED |
| 2 | Wiring harness cable asembly inside of lamp cable. | 2.8.19 | Pankaj Patil | Closed |
| 3 | harness,identification tag bind on both side by black tape | 2.8.19. | Pankaj Patil | CLOSED |
| 4 | PFMEA updation | 10.08.19 | Pankaj Patil | CLOSED |
| 5 | Control Plan Updated | 10.08.19 | Pankaj Patil | CLOSED |
| 6 | SOS Updation | 25.07.19 | Abhishek Sonawane | CLOSED |
| 7 | On job Training given to stage associates for Garnish Fitment . | 2.07.19 | Abhishek Sonawane | Closed |

# Check & confirm the results & effects (Cont……)

## 7.3 Tangible & Intangible benefits:

### Tangible Benefits -

1) Elimination of concern water leakage through Garnish .
2) Rework cost save up to 50940 Rs / Year.
3) Shower FRC improved

### In tangible Benefits

1) Improved Customer Satisfaction.
2) Operator fatigue reduced , moral improved

**From:** DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
**Sent:** Monday, February 10, 2020 4:41:28 PM
**To:** SONAWANE ABHISHEK <SONAWANE.ABHISHEK@mahindra.com>
**Subject:** RE: Sir Please approve

Okay Approved

**From:** SONAWANE ABHISHEK
**Sent:** Monday, February 10, 2020 3:23 PM
**To:** DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
**Subject:** Sir Please approve

| Rework Cost Estimation | | |
|---|---|---|
| Description | Units | |
| Rewok Time | Min | 20 |
| Avg Concerns/Day | Nos | 1 |
| Total Required Rework Time pe | Min | 20 |
| Effective Working time | MIn | 465 |
| Operator Require Per Day | Nos | 0.11 |
| Approximate Cost Per Day | Rs | 1390 |
| Operator Cost/ Concem | Rs | 101.88 |
| Avg no of working days | Days | 25 |
| Operator Cost Per Month | Rs | 2547.1 |
| Operator Cost /Year | Rs | 50,940 |

# 8. Standardization

## 8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

| S.N | Action Taken | Actual date | Responsibility | Status |
|---|---|---|---|---|
| 1 | Control plan updation | 20.08.19 | Pankaj Patil | CLOSED |
| 2 | Process audit check sheet updation | 20.09.19 | Sachin Bhamare | CLOSED |
| 3 | SOS Updation | 25.09.19 | Ganesh Bendale | CLOSED |
| 4 | On job Training given to stage associates for Garnish Fitment associates | 25.06.19 | Ganesh Bendale | Closed |

## 8.2 Training:

| SN | Training Identified | Target date | Responsibility | Status |
|---|---|---|---|---|
| 1. | On job Training given to stage associates | 25.06.19 | Ganesh Bendale | Closed |

## 8.3 Additional Audit & check :

| SN | Document | Target date | Responsibility | Status |
|---|---|---|---|---|
| 1. | Daily Process audit sheet updated | 25.09.19 | Ganesh Bendale | Closed |
| 2 | As per ILU plan, Stage Associate audit done, for adherence to key-point-No NC found | 15.10.19 | Ganesh Bendale | Closed |

# 8. Standardization

## 8.1 Updation of documents :-

### PFMEA Updation -SUPPLIER

**PROCESS FAILURE MODE AND EFFECT ANALYSIS**

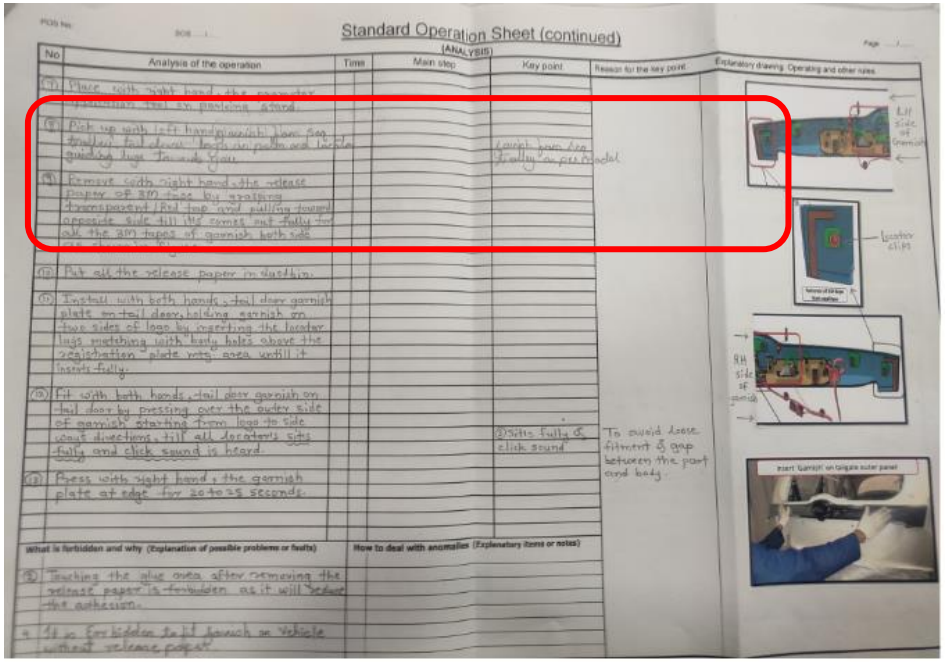
Mahindra

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Part No: | **0108GU300320N/ 330N** | | | Change Letter No.: **NR** | | | | | | System/Subsystem: | | | | | | | | | |
| Part Name: | **U321- Tail Gate Applique (HI/ HI PLUS)** | | | EN No. : **NA** | | | | | | FMEA No. :515 | | | | | | | | | |
| Supplier: | **Exotech Zanini Ind.Pvt.Ltd** | | | Process Responsibility : **Dinu bhokare** | | | | | | Prepared by : **AP** | | | | (Rev.) : | 6 | | | | |
| Vendor Code: | **DE032A** | | | | | | | | | FMEA Date (Orig.) : 10.05.2017 | | | | (Rev. Date) : | **10.08.2019.** | | | | |
| Core Team : | AP / SD / DB / RP / NPD / PP/SK | | | Key Date : **10.05.2017** | | | | | | | | | | | | | | | |
| **Op No.** | **Process Function** | **Requirements** | **Potential Failure Mode** | **Potential Effect(s) of Failure** | **SEV** | **Class** | **Potential Cause(s)/ Mechanisms of Failure** | **Occ** | **Current Process Controls Prevention** | **Current Process Controls Detection** | **Det** | **RPN** | **Recomme-nded Actions** | **Responsibility & Target Comp. Date** | **Actions Taken** | **SEV** | **OCC** | **Det** | **RPN** |
| 270 | Assembly of Substrate ( Wiring harness assembly and bracket) with Applique | 1) Assembly should be proper as per MAster sample.<br>2) Bracket should be as per reference sample.<br>3) Wiring harness assembly as per reference sample. | 1) Improper assebly.<br>2) Wrong bracket assemble..<br>3) Wiring harness cable should be assemble inside lamp cable.<br>4) Wiring harness identification tag bind on bith side. | OS: Nil<br>NU/NP: Part rejected at PDI<br>CU: Part rejected<br>EU: Dissatisfied | 4 | | 1) Assembly procedure not available.<br>2) No identification on WIP box of bracket Hi and Hi plus model in assembly stage,causing it mixed and wrong bracket assembled.<br>3)Assembly operator not followed SOP.<br>4) Received wiring harness from supplier having lable not bind on both side. | 3 | WI for Assembly. | Assy daily production report (F/EMA/01) & rejection register (F/EMA/05) | 8 | **96** | 1) Awareness to be given to assembly operator that wiring harness cable to be assemble inside to lamp cable same OPL ,before after image displayed on workstation.Updat in SOP and Firewall Quality plan.<br>2) Check point for wiring harness identification tag bind on both side to be added in assembly SOP and Firewall QP. | Productionand Quality | 05.08.2019 | 4 | 2 | 8 | 64 |
| | | No abnormal noise after assembly of subsrtate with applique due to extra screw available and fit in bracket. | Abnormal noise during fitment on vehicle. | OS: Nil<br>NU/NP: Part rejected at PDI<br>CU: Part rejected<br>EU: Dissatisfied | 4 | | Assembly operator take more screw in hand during assembly ,causing screw may drop and fit in bracket during screwing in part | 3 | WI for Assembly.and Quality plan | Assy daily production report (F/EMA/01) & rejection register (F/EMA/05) | 8 | **96** | 1) Assembly operator take one by one screw in hand during assembly and same screwing in part to avoid extra screw drop and fit in bracket. 2) Check point for abnormal noise in tailgate Applique added in Quality plan and assembly work instruction. | Production | Action has completed | 4 | 2 | 8 | 64 |

# 8. Standardization

## 8.1 Updation of documents :-

### SOS Updation-TCF

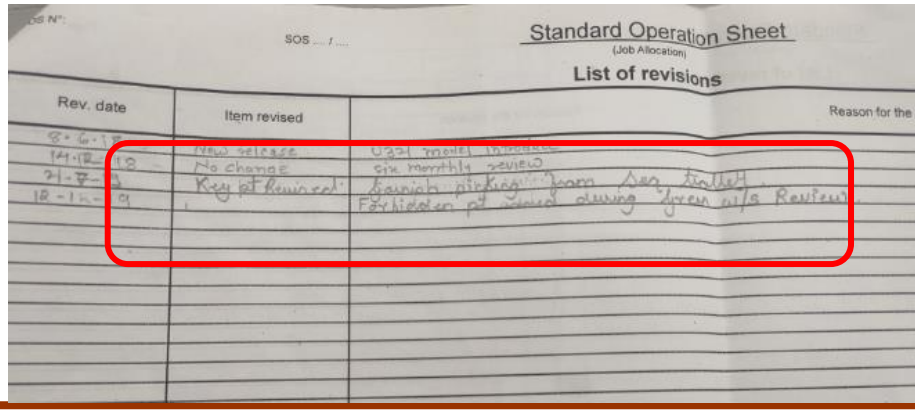


Standard Operation Sheet (continued)

(ANALYSIS)

| No | Analysis of the operation | Time | Main step | Key point | Reason for the key point | Explanatory drawing. Operating and other notes |
|---|---|---|---|---|---|---|
| (7) | Place with right hand, the operator operation tool on parking stand. | | | | | |
| (8) | Pick up with left hand garnish, have gap finisher tool down left in palm and hold guiding lugs towards you. | | | Garnish have fitting as per Model | | |
| (9) | Remove with right hand, the release paper of 3M tape by grasping transparent/Red tab and pulling towards opposite side till 3M comes out fully for all the 3M tapes of garnish both side | | | | | |
| (10) | Put all the release paper in dustbin. | | | | | |
| (11) | Install with both hands, tail door garnish plate on tail door, holding garnish on two sides of logo by inserting the locator lugs matching with body holes above the registration plate mtg area, untill it inserts fully. | | | | | |
| (12) | Fit with both hands, tail door garnish on tail door by pressing over the outer side of garnish starting from logo to side cout directions, till all locators sits fully and click sound is heard. | | | 2) Sits fully & click sound | To avoid loose fitment & gap between the part and body. | |
| (13) | Press with right hand, the garnish plate at edge for 20 to 25 seconds. | | | | | |

| What is forbidden and why (Explanation of possible problems or faults) | How to deal with anomalies (Explanatory items or notes) |
|---|---|
| (1) Touching the glue area after removing the release paper is forbidden as it will reduce the adhesion. | |
| 2 It is forbidden to lift launch an vehicle without release paper. | |



Standard Operation Sheet

(Job Allocation)

List of revisions

| Rev. date | Item revised | Reason for the |
|---|---|---|
| 8-6-18 | new release | U321 model introduce |
| 14-12-18 | No change | six monthly review |
| 21-8-19 | Key pt removed | Garnish picking from Seq. trolley |
| 18-11-19 | | Forbidden pt removed during green m/c Review |

## 8.1 Updation of documents : -

### CONTROL PLAN:SUPPLIER

# CONTROL PLAN

Mahindra

| | | | | | |
|---|---|---|---|---|---|
| **Prototype** | Pre launch | Production | **Key Contact / Phone:** Mr. Dinu Bhokare | **Date(Orig.):** 10.05.2017 | **Date(Rev.): 10.08.2019** |
| **Control Plan Number** : 515 | | | | | |
| **Part Number/LatestChangelevel:** 0108GU300320N/ 330N | | | **Core Team:** NPD/SNK/ SD/PP/DB/AN/AW | **Customer Engineering Approval/Date( if Reqd.):** | |
| Part Name/Description: Lift Gate Applique U321 | | | **Supplier/Plant Approval Date:** | **Customer Quantity Approval/Date( if Reqd.):** | |
| **Supplier/Plant:** Exotch Zanini Ind. Pvt. Ltd. | | **Supplier Code:** DE032A | **Other Approval Date( if reqd.)** | **Other Approval Date( if reqd.)** | |

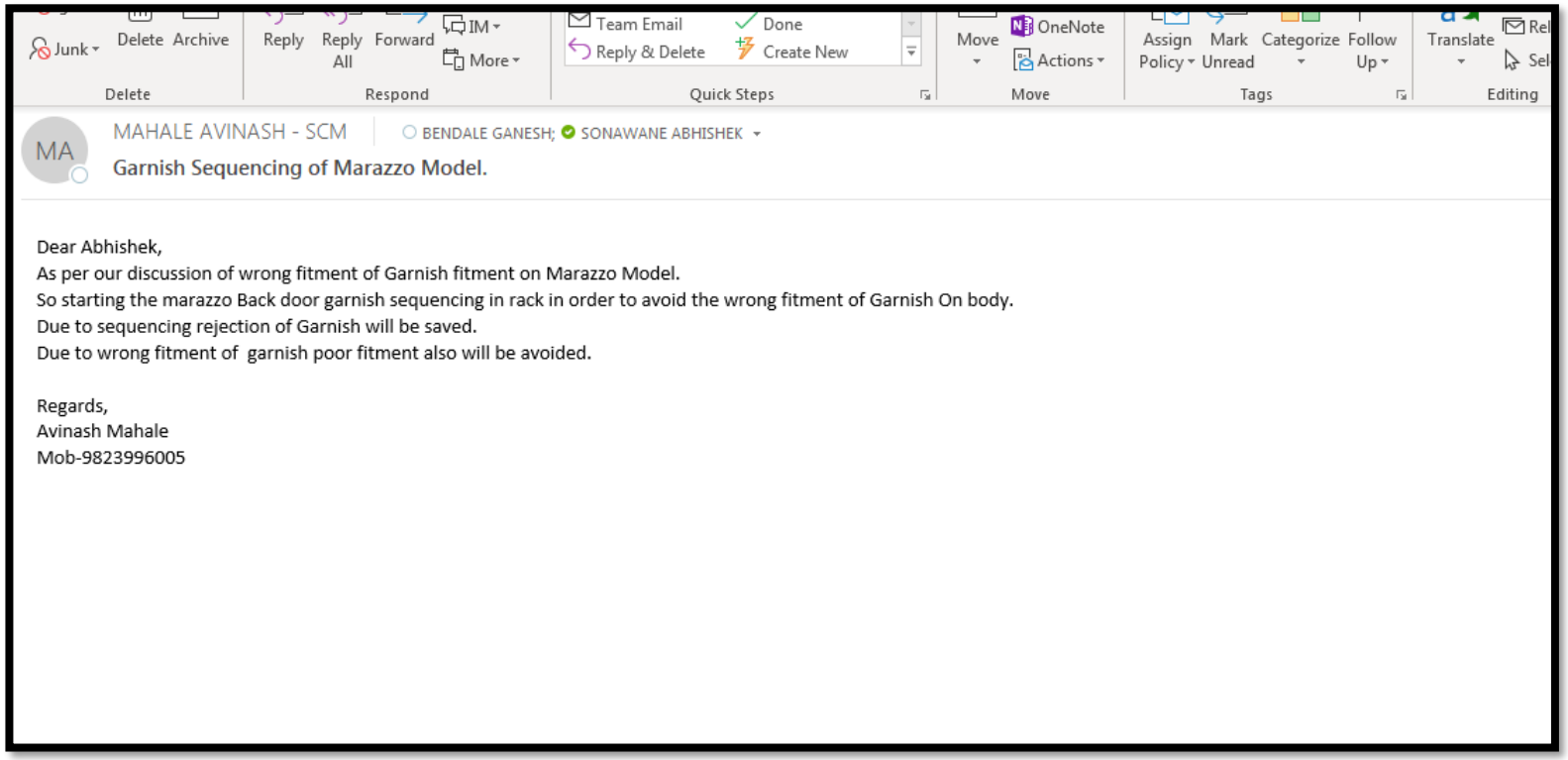
| Part / Process No | Process Name/ Operation Description | M/C, Device, Jig, Tools for Mfg | Characterstics | | | Special Char./ Class | Product / Process Specification/ Tolerance | Evaluation/ Measurement Techniques | Sample | | Control Method | Reaction Plan | Corrective Action |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | NO. | Product | Process | | | | Size | Frequency | | | |
| 270 | Assembly of Substrate (wiring harness and bracket) with Applique | Assly. Fixture | | 1) Assembly should be as per reference sample.<br>2) No abnormal noise after assembly.<br>3) Wrong bracket should not be assembled.<br>4) Wiring harness assemble inside of lamp cable.<br>5) Lamp cable identification lable bind on bothside. | As per Work instruction. | | As per Quality plan. | Visual Inspection | 100% | Every Despatch | Assy daily production report (F/EMA/01) & rejection register (F/EMA/05) | Inform to Supervisor | As per Corrective & preventive action report |

Page 4

# 8. Standardization

## 8.1 Updation of documents : -

### BACK DOOR GARNISH SEQUENCIAL FEEDING COMMUNICATION :



MAHALE AVINASH - SCM | BENDALE GANESH; SONAWANE ABHISHEK

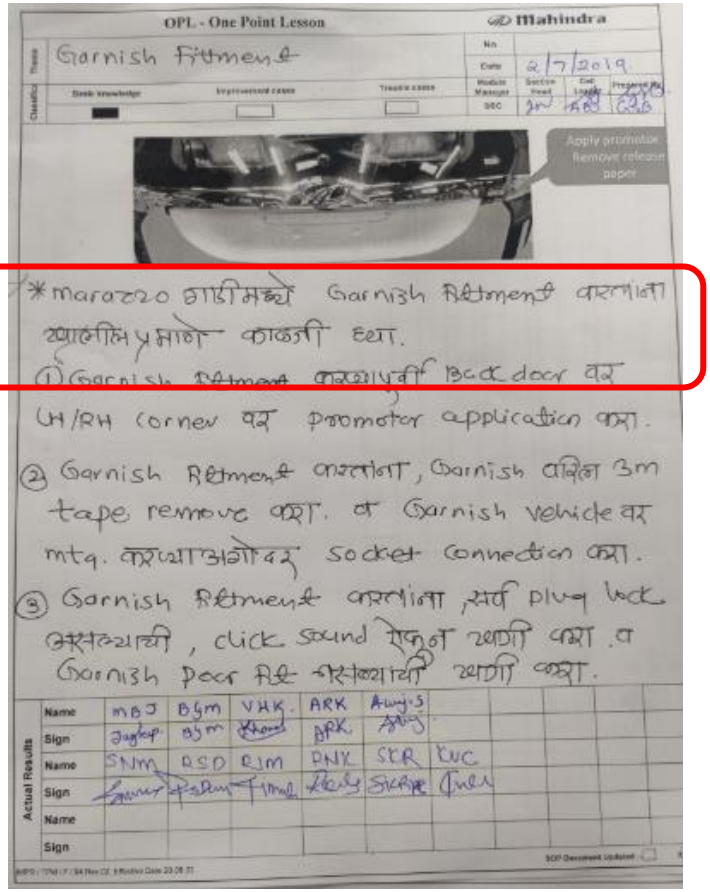
Garnish Sequencing of Marazzo Model.

Dear Abhishek,
As per our discussion of wrong fitment of Garnish fitment on Marazzo Model.
So starting the marazzo Back door garnish sequencing in rack in order to avoid the wrong fitment of Garnish On body.
Due to sequencing rejection of Garnish will be saved.
Due to wrong fitment of garnish poor fitment also will be avoided.

Regards,
Avinash Mahale
Mob-9823996005

## 8.2. Training:

**OPL given to stage associate**



OPL - One Point Lesson

Mahindra

Garnish Fitment

Date 2/7/2019

Apply promotor
Remove release paper

*Marazzo गाडीमध्ये Garnish Fitment करताना खालील प्रमाणे काळजी घ्या.

① Garnish Fitment करण्यापूर्वी Back door वर LH/RH corner वर promotor application करा.

② Garnish Fitment करताना, Garnish वरील 3m tape remove करा. व Garnish vehicle वर mtg. करण्याअगोदर socket connection करा.

③ Garnish Fitment करताना, सर्व plug lock असल्याची, click sound येऊन खाणी करा. व Garnish Door fit असल्याची खाणी करा.

| Actual Results | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Name | MBJ | BGM | VHK | ARK | Anuj.S | | | | |
| Sign | Jagdeep | BGM | Khond | ARK | Anuj | | | | |
| Name | SNM | RSD | RJM | PNK | SCR | KUC | | | |
| Sign | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | | | |
| Name | | | | | | | | | |
| Sign | | | | | | | | | |

# 8. Standardization

## 8.3 Additional Audit & check

### Daily Process audit check sheet updation :-

Mahindra Inspection Standard - Quality

| MIS-Q | Supplier Name | Supplier Code | PAGE |
|---|---|---|---|
| | Exotech Zanini industries Pvt Ltd | DE-032A | 1 |

| Sr.no | PART NAME | Drawing No. | PART NUMBER | Rev No./Date |
|---|---|---|---|---|
| 1 | Tailgate Applique Assembly U321 | 0109GU300320N / High | 0109GU300320N / High | NR/21.05.2017 |
| | | 0108GU300330N / High Plus | 0108GU300330N / High Plus | NR/21.05.2017 |

| S.NO. | INSPECTION ITEM | CRITERIA | INSPECTION METHOD | INSPECTION TOOL | SAMPLING PLAN | INSPECTION STAGE | REMARKS |
|---|---|---|---|---|---|---|---|
| A | FITMENT | | | | | | |
| 27 | Wiring harness cable | Wiring harness cable asembly inside of lamp cable. | Check | Marking | 100% | Final inspection | ..... |
| 28 | Wirig harness identification lable. | Wiring harness identification lable should be bind on both side. | Check | Visual | 100% | Final inspection | ..... |
| B | FINISH | | | | | | |
| 1 | Scratches on the plated/molded surface | Should be free from scratches | Check | Visual | 100% | Final inspection | ..... |
| 2 | Dent Marks | Should be free from Dent | Check | Visual | 100% | Final inspection | ..... |
| 3 | Pit Marks | Should be free from Pit marks | Check | Visual | 100% | Final inspection | ..... |

**All variable parameters should be recorded**
**All functional parameters should have traceability to process control records.**
**Process capability report should be submitted at monthly frequency.**
**Endurance test reports to be submitted as per frequency**

Origin Date : 01.08.2018 Rev No : 7 Rev Date: 10.08.2019.

**For M&M use only :**

**MIS-Q Sr No : 515** **Prepared By : AAJ** **Approved By :**

| Form No / REV No: F/QA/03 | Rev : 0 | Date : | 25.12.10 |
|---|---|---|---|

Sample Size : *Inspection as per* ***Sampling Plan F-QA-03***

# 9. Summarize & Plan future actions

## 9.1 Conclusion:

Water leakage through Back Door Garnish Concern eliminated after taking actions on Wire routing correction, Identification label position correction and Sequential feeding

## 9.2 Horizontal Deployment:

For Upcoming New Model W501

## 9.3 Lessons Learnt:

**Things Gone Wrong** : Failure mode of wire routing incorrect position not identified at time implementation.

**Things Gone Right** : Understood the importance of Wire routing and Identification Label position its impacts and its mandatory consideration in process.

# Thank you

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